Magic states and the experimental road to fault-tolerant quantum computing

Briefing notes for the CONICET interview: connection to OE2 ("Defectos topológicos y medidas de magia cuántica") — V. Benedetti

1. Where quantum computing stands (2025–26)

NISQ is not the goal. Eisert & Preskill [P] stress that today's processors (trapped ions, superconducting circuits, neutral atoms) run circuits with ∼100 qubits and fewer than 10^4 two-qubit operations, with two-qubit errors of 0.1–0.5%. Tasks that are hard to emulate classically exist, but no practically useful computation has been shown. Error mitigation without error correction has a sampling cost that grows exponentially with circuit volume, so it cannot reach deep circuits. The goal is **FASQ** (fault-tolerant application-scale quantum) machines that execute $\sim 10^{12}$ logical operations (a "teraquop"), passing through the megaquop (10^6) and gigaquop (10^9) regimes.

Error correction now works below threshold. Google's surface-code memory reduces the logical error per round by a factor $\Lambda \approx 2$ each time the code distance grows from $3 \to 5 \to 7$ [P]. Harvard's 448-atom processor reaches 2.14(13)× below-threshold performance in a $d = 5$ surface code [H]. Atomic platforms have run logical circuits with 48 logical qubits (280 atoms) and 12 logical qubits (56 ions) [P]. These demonstrations still use few syndrome rounds and heavy post-selection, which does not scale [P].

The cost is enormous. The surface-code estimate in [P] is $P_L \approx 0.1(p/p_{th})^{(d+1)/2}$, with $p_{th} \approx 10^{-2}$. At $p = 10^{-3}$, reaching $P_L = 10^{-11}$ for 10^3 logical qubits needs $d \approx 19$. That means about 10^3 physical qubits per logical qubit, or $\sim 10^6$ in total. Among the hard parts, the **non-Clifford (magic) resource** is widely regarded as the main bottleneck.

2. Why magic is the bottleneck

Stabilizer states and Clifford gates can be simulated efficiently on a classical computer (Gottesman–Knill), so a universal and classically hard computer needs a non-Clifford ingredient. The Eastin–Knill theorem rules out a universal set of *transversal* (naturally fault-tolerant) gates in any code [G, H]. Most codes (surface code, 2D color/Steane code) implement Clifford gates cheaply. The standard fix is **magic-state gate teleportation**: prepare a resource state such as $|T\rangle = (|0\rangle + e^{i\pi/4}|1\rangle)/\sqrt{2}$ offline, then consume it with Clifford operations, Pauli measurements and feed-forward to apply a T gate [G]. Magic states therefore become a manufactured commodity ("magic-state factories"). Their required quality is extreme: infidelity $\sim 10^{-10}$ to factor RSA-2048, and 10^{-6}–10^{-8} for some simulation and optimization tasks [Go].

Zheng, Zang & Kubica [G] give a structural theory of such "magic gate teleportation" (MGT). After back-propagating the Pauli measurements, every MGT protocol is equivalent to encoding the input into a stabilizer code followed by a logical non-Clifford gate. Conversely, within a broad class that includes all known protocols, only resource states that are **Clifford-equivalent to diagonal states** are useful. For one qubit these lie on three great circles of the Bloch sphere. The (1,1,1)-type state $|F\rangle$ from the 5-to-1 distillation protocol is *not* usable for MGT, even though it is a non-stabilizer state. Hence **"having magic" is not the same as being operationally useful magic**, a distinction resource-theory monotones do not see. (Lucas also points to Zhou–Leung–Chuang, quant-ph/0002039, as the classic gate-teleportation reference; it is not among the papers read here.)

3. Experimental state of the art in logical magic states

Group / year	Platform	Method	Encoded state	Result	Accept.
IBM (Gupta et al.) 2023 [I]	Superconducting, heavy-hex, 27-qubit Falcon	Flag-FT measurement of a transversal logical CZ in the [[4,2,2]] error-detecting code; dynamic circuits	$\|CZ\rangle = (\|00\rangle+\|01\rangle+\|10\rangle)/\sqrt{3}$	Infidelity $1.23(11)\times10^{-2}$ (1.87×10^{-2} with feed-forward) vs. best physical 2.4×10^{-2}: **beyond break-even**	∼15–20%
QuEra + Harvard (Rodriguez et al.) 2024 [Q]	Neutral atoms (Rydberg), reconfigurable	**Logical 5-to-1 distillation**: 5 noisy encoded inputs in $d = 3$ and $d = 5$ color codes (85 atoms for $d = 5$), transversal Cliffords	(1,1,1)-direction state	F: 95.1% → 99.4% ($d = 3$), 92.5% → 98.6% ($d = 5$); infidelity suppressed ∼8× and ∼6×	low (post-selected)
Harvard (Bluvstein et al.) 2025 [H]	Neutral atoms, up to 448	Transversal T in 3D [[15,1,3]] Reed–Muller code + teleportation (code switching to Steane); arbitrary-angle synthesis	$\|T_L\rangle$, sequences of up to 3 T's	Angles match theory; spacing shrinks exponentially with T-count (Solovay–Kitaev); error-corrected CHSH = 1.99(3)·√2	varies
Quantinuum (Daguerre et al.) 2025 [D]	Trapped ions, H2-1, 28 qubits	**Code switching** [[15,1,3]] qRM → [[7,1,3]] Steane; verified pre-selection; two-copy Bell-basis certification	$\|T_L\rangle$	Infidelity ≤ $5.1(2.7)\times10^{-4}$, below physical 2q error (1.05×10^{-3}) and SPAM: **below pseudo-threshold**	82.6%
Google (Rosenfeld et al.) 2025 [Go]	Superconducting, Willow	**Magic-state cultivation** in $d = 3$ color code; "kickback tomography"; grafting into $d = 5$ surface code	$\|T\rangle$ (Y-rotated)	Fidelity 0.9999(1): 40× lower error than injection; below every one of 216 physical 2q gates	8%

Lucas's paper [D, Table I] also lists earlier work (2020–24: Bacon–Shor, Steane and surface-code injections on ions and superconducting qubits) with fidelities of 0.88–0.999. Fidelities in different rows were certified differently (logical tomography, two-copy bounds, kickback tomography), with or without post-selection, so they are not strictly comparable.

Four competing routes, one message. (i) *Injection + distillation* [Q]: flexible and code-independent, but costly. (ii) *Error-suppressed/flag encoding* [I]: cheaper, better inputs for distillation. (iii) *Codes with a transversal T + code switching/teleportation* [D, H]: the 3D color code has a transversal T, the 2D one has transversal Cliffords, and teleporting between them evades Eastin–Knill (Harvard shows that code switching and logical teleportation are equivalent). (iv) *Cultivation* [Go]: grows the state inside one small code by repeated fault-tolerant measurement of H_L, with post-selection. All four now produce logical magic states that are **better than the physical operations used to make them**. With Quantinuum's fault-tolerant Clifford gates, this completes a universal logical gate set whose errors are all at or below the physical two-qubit error [D]. Extrapolations: one round of 15-to-1 distillation on Quantinuum's states would give $\sim 4.6\times10^{-9}$ [D]. Cultivation infidelity scales as p^3, so halving p gives $\sim 2\times10^{-5}$ [Go].

Open issues. (a) Heavy reliance on post- or pre-selection, with acceptance falling exponentially at larger distance [D, P]. (b) *Certification*: standard tomography bounds infidelity only as $O(1/\sqrt{N})$, so an error of 10^{-4} needs $\sim 10^8$ shots. Coherent noise can even produce apparent Bloch vectors with norm > 1 [Go]. Hence new estimators: two-copy Bell/SWAP-type purity measurements [D] and kickback tomography [Go], both scaling as O(1/N). (c) Moving the state into a larger, matchable code (grafting) [Go]. (d) Real-time feed-forward, decoding speed and logical clock rate, especially for atoms [P, H].

4. Relation to OE2 (magic measures, topological defects, QFT)

- **Magic is the resource hardware pays for.** OE2 defines magic as distance from stabilizer states, the resource behind classical non-simulability and needed for universality. The experiments show this is not academic. Clifford operations are now “cheap”, and the whole engineering effort (distillation, cultivation, code switching) exists to make non-stabilizer states. Magic monotones do not increase under Clifford operations. The magic of a target state (e.g., a ground state to be simulated) therefore sets a floor on how many magic states, and how large a factory, a fault-tolerant simulation must spend.
- **Magic and entanglement are distinct but intertwined.** OE2 argues this (stabilizer states have a flat entanglement spectrum, so a non-flat spectrum bounds magic from below). Harvard sees it experimentally [H]: logical magic *requires* physical entanglement. $|T_L\rangle$ is an eigenstate of $(X_1X_2\ldots + Y_1Y_2\ldots)/\sqrt{2}$, a superposition across the whole code block. The Clifford decoding circuit conserves magic, so of 15 physical T's applied transversally only one unit of magic survives, which is possible only with entanglement. Unlike Pauli states, T-basis outputs need all of the code's entanglement protected (the error-corrected Bell test).
- **Stabilizer-entropy-type measures are experimentally accessible.** Harvard measured the *additive Bell magic* of Haug & Kim on two copies of a [[15,1,3]] block [H, ED Fig. 9b]. It shows a plateau at exactly one unit of magic under a global T, the same whether the 15 physical qubits or the error-detected logical qubit are analyzed. Bell magic comes from Bell measurements on two copies, much like the stabilizer Rényi entropies of OE2 (Pauli-weighted multi-copy quantities). Quantinuum's certification [D] is also a two-copy (purity/Rényi-2-type) measurement. The replica-trick objects we compute in CFT, with n copies and inserted projectors, are what experiments measure with multiple copies.
- **Topological codes are where stabilizer states live.** The codes in all these experiments (surface, 2D/3D color, Reed–Muller) are lattice topological models whose code states are stabilizer states with flat entanglement spectra. This matches OE2's remark that spin chains and topological theories admit stabilizer states, while in continuum QFT with local degrees of freedom (type III_1 algebras) every physical state has magic. Logical gates in these codes act through the topological structure: transversal T only in the 3D code, Cliffords in 2D, and code switching between them. *My interpretive remark:* this is the natural bridge to the topological-defect language of OE1/OE2, where logical operations act like symmetry defects of the underlying topological phase.
- **Which magic matters?** The MGT result [G] shows that some non-stabilizer states cannot be consumed for gates in the standard (information-hiding) way. This motivates OE2's question of which magic measure is physically meaningful, and suggests pairing monotones such as SREs with operational criteria.
- **Why QFT/criticality?** Preskill notes that early fault-tolerant (megaquop–gigaquop) machines might first reach useful advantage in simulating strongly coupled 2D lattice systems. About 18% of DOE supercomputing cycles go to lattice QCD [P]. Computing the magic of critical/CFT states (e.g., SREs at the Ising critical point via topological defects, and beyond) quantifies the non-Clifford cost of preparing exactly the states such simulations target.

One-paragraph answer for the interview. Quantum hardware has crossed from NISQ into early fault tolerance. Error correction now works below threshold on superconducting, neutral-atom and trapped-ion platforms, and Clifford logic is cheap. The bottleneck is the non-Clifford resource: magic states, consumed through gate teleportation. Since 2023, experiments at IBM, QuEra/Harvard, Quantinuum and Google have produced logical magic states better than the physical operations that made them, reaching infidelities of $\sim 10^{-4}$, using distillation, code switching and cultivation. OE2 studies the theory of this resource: how to quantify magic, how it relates to entanglement (which the experiments show is required for logical magic), and how to compute measures such as stabilizer Rényi entropies in QFT with topological defects and the replica trick. These are multi-copy quantities that experiments already measure.

Sources (folder papers only): [P] Eisert & Preskill, arXiv:2510.19928. [Go] Rosenfeld et al. (Google), arXiv:2512.13908. [Q] Sales Rodriguez et al. (QuEra), arXiv:2412.15165. [H] Bluvstein et al. (Harvard), arXiv:2506.20661. [I] Gupta et al. (IBM), arXiv:2305.13581. [D] Daguerre et al. (Quantinuum), arXiv:2506.14169. [G] Zheng, Zang & Kubica, arXiv:2607.08508. OE2 statements are from the CIC 2025 research plan.